ternational law governing the exploration and use of outer space."[13]

In respect of this treaty, the Soviet government recorded the launch with the Secretary-General, declaring its purpose as "[i]nvestigation of the upper atmosphere and outer space."[14] However, *Kosmos 954* and its sister satellites were, in fact, designed for surveillance of marine vessels.[15] This role inevitably made them subject to potential accidents. In order to monitor ships they sought to bounce active radar signals off the target vessel; to do this effectively they needed to remain in a low orbit, as moving twice as far away from the target reduces the strength of such signals to a sixteenth of their initial intensity. Therefore, the satellites were placed only 150 miles from the surface of the planet, just above the point where atmospheric drag would prevent orbit from being sustained.[16] At such a low altitude there is an increased risk of the craft re-entering the atmosphere should it encounter a problem. Moreover, in low orbits, solar cells cannot be used as a power source, as they would create too much drag, therefore a small nuclear reactor powered the satellite.[17]

The Soviet Union perceived other advantages in utilising a nuclear power source, specifically it increased the operational efficiency of the satellite and improved the weight and size characteristics.[18] Yet these advantages were accompanied by the inherent risk of atomic energy, should an accident occur. It is important to consider that in orbit a nuclear power source does not pose a significant radiation threat. There is no organic life in space. Further, the Sun itself produces large amounts of radiation. The danger is posed either through an accident at

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[13] *Id.* at Preamble.

[14] Information Furnished in Conformity with GA Res 1721 B (XVI), *supra* note 9.

[15] John Lawrence, *Nuclear Power Source in Space a Historical Review*, NUCLEAR NEWS (Nov. 1991).

[16] *Return of the native. (space junk)*, THE ECONOMIST (Oct. 1, 1988), http://findarticles.com/p/articles/mi_hb5037/is_/ai_n18322503.

[17] *Id.*

[18] These aspects were explained by the Soviet delegate Boris Maiorski to the U.N. Committee on the Peaceful Uses of Outer Space. *See* Eilene Galloway, *Nuclear Powered Satellites: The U.S.S.R. Cosmos 954 and the Canadian Claim*, 12(3) AKRON L. REV. 406 (Winter 1979).

launch or due to re-entry into the eco-system. Due to such potential dangers, at the end of their operational life time the *Kosmos* surveillance satellites were raised to a higher orbit where there was little danger of re-entry.[19] *Kosmos 954* was intended to reach an altitude of 900km-1,000km[20] which would have allowed its power source 1,000 years to decay.[21] Although the satellite would still be orbiting the Earth, the Soviet belief was that it would be a sufficient distance away that it would never pose a problem to human activity.

The *Kosmos* satellites were not unique in utilising atomic energy, other satellites have utilised nuclear power sources through radioisotope thermoelectric generators (RTGs).[22] In order to create power, such devices covert the heat generated by natural radioactive decay into electricity.[23] The first such nuclear powered American satellite *Transit 4A* was launched in 1961, its power source was called 'System for Nuclear Auxiliary Power'[24] (SNAP-3), it only produced 2.7 watts but it continued to do so for fifteen years.[25] The first nuclear reactor in space powered the American *OPS 4682* spacecraft in 1965 (the vessel was known as *SNAPSHOT*).[26] As with a terrestrial nuclear reactor, the satellite's power source generated energy through bombarding uranium-235, in order to create a fission reaction.[27] By the time that *Kosmos 954* had completed its operational life time it was estimated to contain 1,000,000 curies of alpha, beta and

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[19] *Cosmos Reentry Spurs Nuclear Waste Debate*, AVIATION WEEK AND SPACE TECHNOLOGY (Jan. 30, 1978).

[20] Nicholas L. Johnson; *A New Look at Nuclear Power Sources and Space Debris*, *in* PROCEEDINGS OF THE 4TH EUROPEAN CONFERENCE ON SPACE DEBRIS 551 (Darmstadt, Germany: European Space Agency, 2005).

[21] NASA Space Link, *Space Nuclear Power Technology*, http://spacelink.nasa.gov/NASA.Projects/Human.Exploration.and.Development.of.Space/Human.Space.Flight/Shuttle/Shuttle.Missions/Flight.031.STS-34/Galileos.Power.Supply/Space.Nuclear.Power.Technology.

[22] Galloway, *supra* note 18, at 404-405.

[23] U.S. Department of Energy, Office of Nuclear Energy, Science, and Technology; *Nuclear Power in Space*, 18-19 (Washington, D.C), www.ne.doe.gov/pdf/npspace.pdf.

[24] The abbreviation "SNAP" is also some times considered to denote "Space Nuclear Auxiliary Programme".

[25] U.S. Department of Energy, *supra* note 23, at 7 & 16.

[26] *A New Look at Nuclear Power Sources and Space Debris*, *supra* note 20.

[27] *Space Nuclear Power Technology*, *supra* note 21.

gamma radiation, roughly the equivalent of a small atomic explosion.[28]

The causes for the crash of *Kosmos 954* are not entirely clear, the Soviet news agency, TASS, reported that the satellite experienced a sharp depressurization on 6th January 1978 due to unknown reasons. It appears probable that this was due to a fuel tank being exhausted, either due to a leak or some form of explosion. This would have preventing the satellite's engines from firing and may have caused a downward thrust towards a lower orbit, and eventual re-entry.[29] A Soviet spokesman, Leonid Sedov, explained this sequence of events as the consequence of *Kosmos 954* being involved in an impact with another craft. If this was the case, it would have been an early instance, perhaps the first, of a satellite being adversely affected by an impact with debris. Such an impact may have prevented any efforts made to control the satellite and prevent its fragmentation.[30]

Raising satellites to a higher orbit, which the USSR intended to use as a means of disposal for the satellite had been successful on other occasions. *Kosmos 952* successfully boosted its orbit to 600 miles above the planet, a 'parking and storage' orbit, on 25th December 1977, as had 14 other surveillance satellites previously.[31] However, when *Kosmos 954* attempted this manoeuvre, a few days after its sister satellite, it began to develop serious problems, which continued into the New Year. On 12th January 1978 United States' President Carter contacted the USSR to offer American assistance in predicting where the troubled craft would crash.[32] However, the Soviets maintained that it would destruct in the atmosphere before it could reach the surface of the planet.[33] The offer of assistance reflected the difficulties associated with such objects returning. When the satellite survived re-entry, less than two weeks after the Ameri-

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[28] Paul G. Dembling, *Cosmos 954 and the Space Treaties*, 6(2) J. OF SPACE L. 131 (1978).

[29] *Id.* at 130.

[30] *Technical Report on Space Debris, supra* note 1, at 93.

[31] Dembling, *supra* note 28, at 129-130.

[32] S.E. Doyle, *Re-entering Space Objects: Facts and Fiction*, 6(2) J. OF SPACE L. 110 (1978).

[33] Galloway, *supra* note 18, at 402.

can approach, NASA/NORAD believed that it would land near Hawaii, whilst the Soviets predicted an area near the Aleutian Islands.[34] During re-entry the majority of the craft did burn up in the atmosphere.[35] The design of the satellite was intended to ensure that the nuclear reactor would completely burn up;[36] however, this did not happen and the Canadian authorities subsequently recovered approximately 0.1% of *Kosmos 954*'s nuclear power source.[37]

THE RECOVERY OPERATION

The initial Canadian efforts to locate the debris, and assess the damage, involved more than 400 scientists and military personnel, along with 12 aircraft;[38] an operation conducted with the assistance of the United States.[39] The US flew KC135 jets and U-2s equipped with filters designed to trap radioactive fallout still in the upper atmosphere.[40] The debris was scattered over an area estimated to be 124,000 square kilometres,[41] as such it is not surprising that the recovery operation lasted for most of 1978.[42]

At launch *Kosmos 954* weighed approximately 5 tons;[43] fragmentation in the upper atmosphere, reduced it to approximately 4,000 pieces.[44] Most importantly, several thousand pieces of debris were later reported to be 1mm spheres, which had cores of nearly pure uranium-235, these were scattered over an area of

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[34] Doyle, *supra* note 32, at 110.

[35] United Press International (Jan. 7, 1983).

[36] *Canada: Claim Against the Union of Soviet Socialist Republics for Damage Caused by Soviet Cosmos 954*, 18 I.L.M. 899, 903 (1979) [hereinafter *Canada: Claim Against the Union of Soviet Socialist Republics*].

[37] *"The Cosmos 954 Accident" Healthy Environments and Consumer Safety*, www.hc-sc.gc.ca/hecs-sesc/neprd/nep-events/cosmos.html (last visited Nov. 13, 2008) [hereinafter *The Cosmos 954 Accident*].

[38] *Radioactive Piece of Soviet Satellite Round*, FACTS ON FILE WORLD NEWS DIGEST (Feb. 17, 1978).

[39] *Id.*

[40] Dembling, *supra* note 28, at 126.

[41] *The Cosmos 954 Accident*, *supra* note 37.

[42] *Canada: Claim Against the Union of Soviet Socialist Republics*, *supra* note 36, at 929.

[43] Dembling, *supra* note 28, at 126.

[44] Note Verbale Dated 19 December 1978, *supra* note 9.

several hundred kilometres.[45] In its claim for compensation, against the U.S.S.R., the Canadian government stated that it considered some of the fragments found to have a lethal level of radioactivity.[46] A lethal dose of radiation is approximately 500 roentgens; however, smaller doses are still harmful, especially when coupled with long term exposure; workers at nuclear installations are not allowed to absorb more than 5 roentgens per year.[47] A single piece of *Kosmos 954*, found in the region of Artillery Lake, a 10cm long metal cylinder, was emitting 100 Roentgens per hour.[48]

By 8th February 1978, the Canadian authorities did not consider that they had received sufficient information from the U.S.S.R. This resulted in the view that their enforced ignorance was hampering the search for remnants of the satellite. Answers as to the nature of *Kosmos 954*'s fuel, the shielding in which it was contained and most importantly how the Soviet Union would have conducted the recovery operation had the satellite landed in their territory, had not been provided.[49]

Canadian translations of the Soviet response notes that the embassy expressed regret that the U.S.S.R. had not participated in the recovery operation in Canada, "guided by the rules and principles of international law." The response asserted that such assistance had been immediately offered when debris was discovered in Canadian territory.[50] Yet the Soviet Union's response did not answer the technical questions which Canada had posed. Canadian communications to the U.S.S.R., on 28th February 1978, emphasised that the assistance they required was not the Soviet Union's physical help in recovering fragments of the satellite, but technical information to aid the search.[51]

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[45] Lawrence, *supra* note 15.

[46] *Canada: Claim Against the Union of Soviet Socialist Republics*, *supra* note 36, at 904.

[47] *Radioactive Piece of Soviet Satellite Round*, *supra* note 38.

[48] Note Verbale Dated 3 March 1978, *supra* note 9.

[49] *Canada: Claim Against the Union of Soviet Socialist Republics*, *supra* note 36, at 913-915.

[50] The Canadian legal documents from which this information is taken notes that they are an unofficial translation of the Soviet Union's statement. *Id.* at 915.

[51] *Id.* at 916-919.

Canada continued to pose technical questions for several months during 1978, the most detailed set being raised on 13th April.[52] The largest problem in the communications concerning the incident appears to have been the nature of *Kosmos 954*. As a 'spy satellite' the U.S.S.R. appeared reluctant to reveal information concerning its constitution, commenting:

> The Soviet side finds it also necessary to note that some of the questions put by the Canadian side obviously relate to the information which is outside of the scope of the amount necessary to secure health and safety of persons and the environment.[53]

However the suggestion that the U.S.S.R. was not forthcoming with assistance concerning *Kosmos 954* was somewhat contradicted by President Carter's Assistant for National Security Affairs, Dr. Z. Brezinski, who stated that the situation "was handled through very effective international cooperation, involving the Soviets also."[54] Further, the Soviet delegate Boris Maiorski specifically reported to the United Nations Committee on the Peaceful Uses of Outer Space (UNCOPUOS), that as soon as it became apparent that *Kosmos 954* had fragmented over Canada, the government of the U.S.S.R. offered immediate assistance.[55] This interpretation of events was disputed by the Canadian delegate, Erik B. Wang, who maintained that although the U.S.S.R. had offered to cooperate, this offer had been subsequent to the search operation commencing, further the information provided concerning the satellite had not been sufficient nor complete.[56]

When the recovery operation was completed, the effects of *Kosmos 954* upon the natural environment were not considered

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[52] *Id.* at 923-926.

[53] The Canadian legal documents from which this information is taken notes that they are an unofficial translation of the Soviet Union's statement. *Id.* at 928.

[54] Galloway, *supra* note 18, at 402.

[55] Maiorski then pointedly commented that "[t]he question of how that aid was used related to an area in which every Government makes independent and sovereign decisions." *Id.* at 406-407.

[56] *Id.* at 407-408.

to be significant.[57] Further, although some local residents were exposed to small amounts of radiation none suffered any serious harm,[58] and there was no detectable contamination of "air, water or food supplies."[59] It was fortunate that the debris had fallen into a largely uninhabited region, rather than a dense urban population. Despite the relative good fortune, regarding this particular incident, there are two important factors to consider; firstly it was not a unique event, nor is there reason to believe that such an impact, or one more serious, could never occur again. Secondly, the cost of repairing the damage entitled Canada to make a formal claim for reparations against the U.S.S.R., and thus exercise liability as it is contained within space law.

OTHER INCIDENTS OF SATELLITES FALLING TO THE EARTH

The crash of *Kosmos 954* was the most dramatic instance of an artificial space object crashing to Earth, and the first to create a crisis,[60] however it was not unique. In September 1962, at a meeting of the UN Outer Space Committee, the United State's delegate presented a twenty pound piece of metal, that had been found in Wisconsin. It was believed to be debris from the Soviet satellite *Sputnik IV*, which had crashed into American territory.[61] The exhibit was an early demonstration of the danger of spacecraft fragments accidentally landing on the surface of the planet; a threat which was partially realised by *Kosmos 954*.

Prior to the *Kosmos 954* incident, the United States had been involved in three occasions in which nuclear power sources returned from space, one burned up in the atmosphere, whilst parts of the other two were recovered from the oceans by the

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[57] Lawrence, *supra* note 15.

[58] James Oberg, *The Probe That Fell To Earth*, NEW SCIENTIST (Mar. 6, 1999).

[59] NASA Space Link, *Space Nuclear Power System Accidents*, http://spacelink.nasa.gov/NASA.Projects/Human.Exploration.and.Development.of.Space/Human.Space.Flight/Shuttle/Shuttle.Missions/Flight.031.STS34/Galileos.Power.Supply/Space.Nuclear.Power.System.Accidents.

[60] Stephen Gorove, *Cosmos 954: Issues of Law and Policy*, 6(2) J. OF SPACE L. 138 (1978).

[61] The Soviet delegate declined to take the object as a gift. BIN CHENG, STUDIES IN INTERNATIONAL SPACE LAW 286-287 (Oxford, Clarendon Press, 1997).

Navy.[62] One of these incidents, in April 1964, concerned the return of the US satellite *Transit 5BN3*; this was powered by *SNAP 9*. It crashed into the Indian Ocean when its launch vehicle failed, and until *Kosmos 954* this was the worst such incident.[63] Subsequent to *Kosmos 954*, in 1979, debris from an American *Skylab* satellite landed in Australia; no governmental claim for damages was made, as there was no clean up cost, however the United States did refuse the claims of approximately 40 Australian citizens who cited mental distress caused by *Skylab*.[64] Similarly, the Russian *Mars-96* probe, dramatically failed to reach Mars, crashing back to Earth and scattering plutonium over Chile. In this instance it was decided to level the debris *in situ* rather than attempt to retrieve it.[65]

Given the dangerous operation that the *Kosmos* surveillance satellites performed, it is not surprising that they encountered further problems. Following the *Kosmos 954* incident the Bouk reactors, which powered the satellites, were redesigned such that the nuclear core would be ejected, in order to ensure that it was completely vaporized in the atmosphere.[66] This new system proved successful when *Kosmos 1402* re-entered the atmosphere, above the Indian Ocean.[67] The core successfully ejected, burning up high above the planet and the radiation was dispersed over a very wide area.[68] *Kosmos 1900* also encountered difficulties, on this occasion, the satellite was also able to eject its nuclear reactor before it re-entered the Earth's atmosphere. The reactor core was sent into a higher orbit, where it posed no foreseeable threat.[69]

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[62] One of the items recovered from the sea was part of the ill fated Apollo 13 mission. *See* Dembling, *supra* note 28, at 131.

[63] David S.F. Portree & Joseph P. Loftus, *Orbital Debris: A Chronology*, NASA/TP-1999-208856, at 6 (Jan. 1999).

[64] Margaret B. Carlson, *Space Law Launches Increasing Number of Lawyers*, LEGAL TIMES (June 14, 1982).

[65] Oberg, *supra* note 58.

[66] Space Nuclear Power Technology, *supra* note 21.

[67] Bryan Brumley, *Nuclear Debris Expected To Rain To Earth This Week*, THE ASSOCIATED PRESS (Oct. 1, 1988).

[68] *Return of the native. (space junk)*, *supra* note 16.

[69] *Spy Satellite Reactor Now in a Safe Orbit, Its Trackers Report*, THE NEW YORK TIMES (Oct. 5, 1988).

The problem of satellites returning into the atmosphere has persisted, although most occasions do not have dramatic consequences. The Soviet satellite *Kosmos 1402* crashed back to the Earth in February 1983,[70] whilst in January 2005 a US craft crashed near Bangkok,[71] and *Kosmos 389* is about to re-enter the atmosphere.[72] Perhaps the most remarkable such instance occurred in January 1995, when after three hours in orbit a Japanese-German satellite *Express* returned to Earth. It crashed in Ghana, where it is reported that tribesmen took it as a gift from the gods and worshipped it accordingly. It was then retrieved by the army who, *The Ghanaian Chronicle* inferred, believed that it was alien in origin. This incident also demonstrated the difficulty in predicting where an object will crash land, the satellite's German operators had expected it to crash into the Pacific Ocean.[73]

When these instances are compared to the danger posed by debris, the increased threat of the new problem can be seen. The majority of re-entries do not result in any serious consequences and few are widely reported. However, the potential of the new threat is such that it could seriously disrupt space operations over a time scale measured in centuries.

REACTION TO THE INCIDENT

Within a week of *Kosmos 954*'s crash President Carter expressed his desire for an agreement with the Soviet Union, whereby nuclear materials would not be used onboard satellites orbiting the Earth. This proposition did not envisage a prohibition upon their use for deep space missions, which operate too

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[70] H. Klinkrad, et al., *Space Debris Activities in Europe*, *in* PROCEEDINGS OF THE 4TH EUROPEAN CONFERENCE ON SPACE DEBRIS 25 (Darmstadt, Germany: European Space Agency, 2005).

[71] Nicholas Johnson, *Orbital Debris Research in the U.S.*, *in* PROCEEDINGS OF THE 4TH EUROPEAN CONFERENCE ON SPACE DEBRIS 9 (Darmstadt, Germany: European Space Agency, 2005).

[72] Sergey Kulik, *The Russian Federation Space Plan 2006-2015 and Activities in Space Debris Problems*, *in* PROCEEDINGS OF THE 4TH EUROPEAN CONFERENCE ON SPACE DEBRIS 15 (Darmstadt, Germany: European Space Agency, 2005).

[73] Chris Oliver Wilson, *Space-Shot Flop Becomes Tribesmen's Fallen Idol*, THE SUNDAY TELEGRAPH 24 (Feb. 18, 1996).

far from the Sun for solar energy to be used effectively.[74] The Canadian government pursued two legal objectives, firstly gaining compensation for the crash, secondly focusing upon safe means by which nuclear power could be used in space.[75]

The accident highlighted the serious dangers of nuclear power sources being utilised in orbit. Before this incident most Nuclear Power Source (NPS) programmes had been largely abandoned as solar cells offered a more efficient means of powering satellites, though the *Russian Radar Ocean Reconnaissance Satellite* (*RORSAT*) programme, of which *Kosmos 954* was part,[76] had continued into the 1970s.[77] The practice has now been completely abandoned; no nuclear powered satellites have been launched since 1988, but fifty three such satellites remain in orbit, largely located at around an altitude of 1,000km. This altitude was chosen as a disposal orbit before space debris was considered to be a serious problem, this is unfortunate as it is now identified as having one of the highest densities of debris.[78] It has been suggested that the nuclear powered satellites which remain in orbit have contributed to high debris densities. Leakages from the coolant systems of these satellites are considered to be the source of particles approximately 1cm in diameter. The total number of such particles, with a combined mass of 30kg, is estimated to be 80,000.[79] Thus, NPS satellites left in orbit, are in a region of a high debris density, and are potentially a source of large numbers of small pieces of debris.

In the decade that followed the cessation of NPS being launched, the U.S.S.R. and U.S.A. conducted considerable research into new nuclear reactors for usage in space.[80] More recently, some states have considered the possibility of again us-

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[74] Galloway, *supra* note 18, at 404-405.

[75] Jason Reiskind, *Towards A Responsible Use of Nuclear Power in Outer Space – The Canadian Initiative In The United Nations,* VI ANNALS OF AIR AND SPACE LAW 463-464 (1981).

[76] Portree and Loftus, *supra* note 63, at 25.

[77] *A New Look at Nuclear Power Sources and Space Debris, supra* note 20.

[78] *Id.* at 552 & 554.

[79] A.I. Nazarenko, et al., *Spacecraft with a Nuclear Power System and Problems of Space Debris, in* PROCEEDINGS OF THE 4TH EUROPEAN CONFERENCE ON SPACE DEBRIS 558 (Darmstadt, Germany: European Space Agency, 2005).

[80] *A New Look at Nuclear Power Sources and Space Debris, supra* note 20.

ing NPS onboard satellites, specifically radioisotope power systems (RPS) and nuclear propulsion systems.[81] There has also been political attention paid towards the subject, in 1992, the United Nations General Assembly highlighted the associated dangers, when it adopted *Principles Relevant to the Use of Nuclear Power Source In Outer Space* (Resolution 47/68).

When considering the danger of NPS in near Earth orbit, it is necessary to note that whilst they remain in space their radioactive nature does not pose any threat to humanity. Should a NPS satellite fragment, the resulting debris would be no more dangerous than any other type of debris, due to the absence of biological life in space. Further, it is projected that debris resulting from a nuclear power source would be expected to be less able to survive re-entry into the atmosphere than conventional debris, thus reducing the risk of radiation being able to enter the eco-system.[82]

The practice of placing defunct satellites with nuclear power sources on board into a relatively high orbit, where they would pose no danger, and the fission fragments could naturally decay, appeared to be an unproblematic solution when it was put into practice, in the 1970s. However, this is now in contradiction with the efforts to preserve near Earth space from the problem of debris. Orbits close to the planet are now conceptualised as a resource in need of protection. As such, leaving nuclear power sources in orbit to decay is in conflict with the Inter-Agency Debris Coordination Committee's (IADC)[83] guidelines, which stipulate that satellites should not remain in LEO for more than 25 years after the end of their operational life time. Nicholas Johnson suggests that this conflict could be resolved by the usage of storage orbits above LEO.[84] As the orbits selected would be used infrequently, it is believed that, the radioactive debris would pose little threat.

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[81] *Id.* at 551.

[82] *Id.* at 553.

[83] The IADC is a body constituted of all the major space faring powers. It proceeds on the basis of cooperation with the objective of creating a framework within which to manage the debris problem.

See Inter-Agency Space Debris Coordination Committee, www.iadc-online.org.

[84] *A New Look at Nuclear Power Sources and Space Debris*, *supra* note 20.

LEGAL FRAMEWORK WITHIN WHICH THE CRASH WAS MANAGED

The 1967 Outer Space Treaty[85] establishes the broad principles upon which the exploration and exploitation of space occurs. Article VI of the treaty places responsibility upon states for their national activities conducted in space, or upon celestial bodies. It continues, in Article VII, to place liability upon a launching state for damage caused to any other state party to the treaty, by an object which it has launched. Article VIII of the treaty also has implications in the case of *Kosmos 954*, it commits states to return, to the state of registry, objects which accidentally land in their territory.

The Rescue Agreement,[86] to which both Canada and the U.S.S.R. were party,[87] re-enforced Article VIII of the Outer Space Treaty. Although primarily concerned with the treatment of personnel in space the Rescue Agreement, in Article 5, addresses objects which return from space. In accordance with the treaty's provisions, Canada informed the U.S.S.R. that component parts of one of its space objects had been located in Canadian territory.[88] Further, with reference to Paragraph 1 of Article 5 of the treaty, Canada informed the Secretary General of the United Nations that the remnants of *Kosmos 954* had been found in its territory.[89]

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[85] Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies, Jan. 27, 1967, 18 U.S.T. 2410, T.I.A.S. No. 6347 [hereinafter Outer Space Treaty]. The U.S.S.R. was a depository government for the treaty, whilst Canada was a party to it from 1967, *See* U.S. Department of State, *Current Treaties and Agreements*, www.state.gov/t/ac/trt/5181.htm.

[86] Agreement on the Rescue of Astronauts the Return of Astronauts and the Return of Objects Launched into Outer Space, Apr. 22, 1968, 19 U.S.T. 7570, U.N. GAOR 22nd Sess., Supp. No. 16, U.N. Doc A/6716 [hereinafter Rescue Agreement].

[87] The U.S.S.R. was one of the depository governments for the treaty (Article XIV), whilst Canada signed it in 1968 and ratified it on 20th February 1975. *See* Agreement on the Rescue of Astronauts (Summary Information), http://pubx.dfait-maeci.gc.ca/A_BRANCH/AES/env_commitments.nsf/0/4ac0c900f6f6d1e985256b6c004aeba7?OpenDocument.

[88] The U.S.S.R. was formally informed on 8th February 1978. *See, Canada: Claim Against the Union of Soviet Socialist Republics*, *supra* note 36, at 910-911.

[89] The United Nations provides a list of objects found which have been reported through their offices. *See*, U.N. Office for Outer Space Affairs, *List of Reported Space Objects Discovered by Member States within their territories*, U.N. Doc. A/AC.105/214 (Feb. 8, 1978), *available at* http://www.unoosa.org/oosa/natact/sdnps/unlfd.html.

Paragraph 3 of Article 5 provides for the return of objects, and their component parts. This requirement had little importance, the objects recovered were radioactive waste, as such the Soviet Union did not express a desire to have them returned. On 20th February 1978 the U.S.S.R. formally notified Canada that it could dispose of the recovered items at its discretion.[90]

The treaty also provides states with a means to request assistance from the launching state, in the recovery of objects which have landed in their territory.[91] Although it had the right to do so Canada did not request physical Soviet assistance in the recovery operation.[92] Paragraph 4, of Article 5, specifically deals with occasions when hazardous materials are found to have landed within the territory of a state. In such circumstances, the launching state is obliged to take immediate actions in order to "eliminate possible danger." It would appear that under the terms of this paragraph, Canada had the authority to request that the U.S.S.R. itself retrieved the material from Canadian territory, then dispose of it elsewhere. Paragraph 5 of the same Article refers to the costs involved in recovering crashed space objects, these are to be born by the launching state, as with a similar general principle contained within the Outer Space Treaty,[93] this notion was giving a more specific legal form in a later treaty designed to specifically address issues of liability.

The Liability Convention[94] does not stand in isolation, it builds upon the principles of the Outer Space Treaty and the Rescue Agreement. The treaty was produced by UNCOPUOS

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[90] The Canadian legal documents from which this information is taken notes that they are an unofficial translation of the Soviet Union's statement. *See, Canada: Claim Against the Union of Soviet Socialist Republics, supra* note 36, at 915-916. The USSR reiterated this position on 21st March 1978. *Id.* at 922-923.

[91] Rescue Agreement, *supra* note 86, at art. 5, para. 2.

[92] Galloway, *supra* note 18, at 411.

[93] Outer Space Treaty, *supra* note 85, at arts. VI & VII.

[94] The U.S.S.R. is one of the three depository governments for the Liability Convention (Article XXIV). The treaty entered into force on 1st September 1972, when it received its fifth ratification (CHENG, *supra* note 61, at 286), Canada acceded to the treaty on 20th February 1975. *See, Convention on International Liability for Damage Caused by Space Objects (Summary Information)*, www.fco.gov.uk/Files/kfile/015_DamageSpaceObjects.pdf.

after nine years of negotiations,[95] the process of completing this treaty was considered by Herbert Reis, a legal advisor to the American Mission to the United Nations, as one of the most difficult in the post-war era.[96] It was this treaty which provided Canada with the primary legal basis upon which it made its claim for compensation, this was not the only option available, had it wished the Vienna Convention on Civil Liability for Nuclear Damage offered an alternative legal recourse.[97]

Ironically, during the drafting of the *Liability Convention*, the U.S.S.R. had argued that 'nuclear damage' should be excluded from the convention, and addressed in a separate treaty. This was virtually unanimously opposed by the other members of the Legal Sub-Committee of UNCOPUOS, and in 1969 the Soviet objection was withdrawn.[98]

The initial articles of the Liability Convention clearly reveal its applicability in this instance, Article I defines a space object as including 'component parts of a space object'. Whilst it continues in Article II stating that, '[a] launching State shall be absolutely liable to pay compensation for damage caused by its space object on the surface of the Earth'. There is little doubt concerning the relevance of this article, as the identity of the object which crashed into Canadian territory, was never disputed. Although there was no loss of life, or serious health implications, resulting from the crash of *Kosmos 954* the damage to Canadian property, specifically its environment, permitted the treaty to be invoked.[99]

Because the U.S.S.R. never denied the radioactive fragments located in Canada were of Soviet origin,[100] the compensation claim proceeded rapidly; Article X of the *Liability Convention* requires that a state makes its claim for compensation

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[95] CHENG, *supra* note 61, at 286. Cheng provides a thorough account of the drafting process and review of the text of the treaty. *Id.* at 286-356.

[96] Herbert Reis, *Some Reflections on the Liability Convention for Outer Space*, 6(2) J. OF SPACE L. 125 (1978).

[97] Gorove, *supra* note 60, at 144.

[98] CHENG, *supra* note 61, at 323-324.

[99] Liability Convention, *supra* note 5, at art. I.

[100] *Canada: Claim Against the Union of Soviet Socialist Republics*, *supra* note 36, at 902-905.

within one year of the damage occurring, or the identification of the responsible party, should that not be immediately known. Due to the latter clause not being relevant Canada was able to act promptly. The legal process was given increased pace and clarity as the USSR made it clear, before Canada made a claim, that it was willing to pay compensation.[101]

Stephen Gorove observes that, although neither term has been formally defined, within space law the terms 'liability' and 'responsibility' have differing meanings although they are closely connected. The former refers to liability which a state bears for damaged caused by objects which it has launched into orbit, whereas the latter refers to the more general international responsibility which a state has for its activities in outer space. Therefore responsibility denotes the norms of behaviour that apply to states and their representatives.[102]

The concept of 'liability' in space law is further complicated because of the potential applicability of international law, domestic law and foreign law.[103] Thus in the case of *Kosmos 954* arguments could have been made that liability should have been addressed under Soviet law, as the satellite originated there, or Canadian law as the damage occurred in its territory.

Within the *Liability Convention* there are differing degrees of liability in accordance with the location in which damage is caused. Liability is limited when an incident occurs in air space, or in outer space, as the treaty states that damage caused "other than on the surface of the Earth" liability is dependent upon their being fault in the actions of a state.[104] This has clear implications for scenarios involving debris, as the damage would be caused in outer space. In order for the treaty to be used successfully, there would be a necessity to prove fault. However, when damage is caused on the surface of the Earth, such as the case of *Kosmos 954*, the highest degree of liability is incurred, covering virtually all damage caused, without restriction.[105] In theory,

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[101] Gorove, *supra* note 60, at 138.
[102] *Id.* at 373-379.
[103] *Id.* at 373.
[104] CHENG, *supra* note 61, at 326-328.
[105] *Id.* at 320-323.

this could have resulted in the U.S.S.R. being liable to pay an enormous amount, however, as the satellite landed in an uninhabited region, there was no loss of life due to the crash nor was there damage to private property. However, because of the nuclear power source on board, the area needed to be decontaminated, the Soviet Union was liable for the cost of this clean up; Canada accordingly made a claim against the U.S.S.R.

The Soviet Union could have pleaded exception from liability under Article VI if it demonstrated that the events were a result of gross negligence by Canada,[106] this would have been a difficult argument as Canada was a fixed target with which the satellite collided. Similarly when considering debris, it is difficult to imagine how an object damaged by debris could be considered to be grossly negligent if it were merely orbiting the Earth.

In consideration of damages the Liability Convention states that they should be paid to restore the claimant "to the condition which would have existed if the damage had not occurred."[107] It is notable that the purpose of the treaty is to grant restorative not punitive damages, which suggests the assumption has been accepted that any damage which occurs will be the product of accidents, rather than belligerent intent. Clearly there would only have been a very weak argument to suggest that the U.S.S.R. should have been subject to punitive damages, as the incident was the product of an accident, although the U.S.S.R. had made the decision to place a nuclear powered satellite in a very low, and therefore hazardous, orbit. With reference to the debris problem, punitive damages would not appear to be appropriate either as a large amount of the current debris population was produced before the problem was identified, further what is still produced is not done so deliberately. However, should explosions occur in orbit due to the use of weapons, then a far stronger philosophical argument would exist for punitive damages to be demanded by a third party who's resources were damaged as a consequence. Yet it is difficult to conceptualise a

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[106] Dembling, *supra* note 28, at 133.
[107] Liability Convention, *supra* note 5, at art. XII.

scenario in which a punitive notion of liability could be introduced into space law.

Canada originally made a claim for C$6million[108] in damages and costs, whilst the total price of the clean up was estimated to be as high as C$14million.[109] The difference between the two amounts, is due to Canada claiming for only reasonable costs and those which it could accurately calculate.[110] Ultimately, in 1981, the Soviet Union agreed to pay C$3million[111] due to the damage caused, however it did not admit liability for the incident.[112]

As the Soviet Union offered an acceptable payment, there was no requirement for the provisions relating to settlement of the disputes to be invoked.[113] Had the two states not reached agreement through diplomatic negotiations within one year of a notification of a claim being made[114] concerning the amount of compensation to be paid, the Liability Convention has provision for a Claims Commission to be established. This body would be composed of three members who would investigate the basis of the claim, one member being appointed by each party whilst the third member, the chair, would be appointed by mutual consent.[115]

In the opinion of Margaret Carlson, had the U.S.S.R. and Canada failed to reach an agreement, the practical limitations of the Liability Convention would have been exposed. The treaty provides for a Claims Commission to be established, however it

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[108] The Canadian claim was for the exact figure of $6,041,174.70 (Canadian). *Canada: Claim Against the Union of Soviet Socialist Republics*, *supra* note 36, at 899.

[109] *Canada-Soviets Sign Satellite Damage Pact*, THE ASSOCIATED PRESS (Apr. 2, 1981).

[110] *Canada: Claim Against the Union of Soviet Socialist Republics*, *supra* note 36, at 906-907.

[111] The sum was in Canadian not American dollars, in U.S. dollars the figure was approximately $2.55million. *See* Andrew Cohen, *Canada Settles With Russia for Satellite Crash*, UNITED PRESS INTERNATIONAL (Apr. 2, 1981).

[112] *Canada-Union of Soviet Socialist Republics: Protocol on Settlement of Canada's Claim for Damage Caused by "Cosmos 954"*, 20 I.L.M. 689 (1981); J.E.S. FAWCETT; OUTER SPACE 26-27 (Oxford, Clarendon, 1984).

[113] *Orbiting Debris, A Space Environmental Problem - Background Paper*, *supra* note 7, at 32-33.

[114] Liability Convention, *supra* note 5, at art. XIV.

[115] *Id.* at arts. XIV – XX.

does not grant it binding authority upon the disputing parties.[116] Bin Cheng notes that should both parties in the arbitration process not agree with the verdict of the Claims Commission it shall reach a final decision, which is not subject to appeal. However, this remains a "recommendatory award," there is no legal mechanism to make the decision of the Commission binding; parties can only be bound by its finding if they chose to be so.[117] "Recommendatory awards" are not without precedent in international law; Advisory Opinions of the International Court of Justice are a similar form of decision.[118] Although they many not appear to be an ideal mechanism for the resolution of a dispute, this should not be considered a particular weakness of the Liability Convention, rather it is a reflection upon an international system in which there is no authority which can enforce decisions upon states.

An alternative position towards disputes between states is adopted within the European Space Agency (ESA) wherein, according to Article XVII of its founded convention, the verdict of a tribunal, established to investigate a dispute, is final and binding upon parties.[119] However, in this instance states have agreed, by virtue of their membership of the organisation, to be bound by the decisions of such tribunals prior to a dispute arising. Further, ESA has the capacity to impose verdicts upon member states because of their close linkage within the Agency. Such a framework would not apply within general space law, as there is no similar close association of states which can be used to enforce verdicts.

The U.S.S.R.'s actions, or lack of action, during the recovery operation did not violate the principles of space law, partially because space law attempts to resolve issues *post hoc*, rather

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[116] Carlson, *supra* note 64.

[117] Liability Convention, *supra* note 5, at art. XIX, para. 2 (provides states with the right to make the findings of the Commission binding).

[118] There was disagreement concerning this structure, in effect of which, Canada, Japan, Iran and Sweden abstained when votes were taken in the First Committee and the General Assembly on votes to approve the draft convention. *See* CHENG, *supra* note 61, at 353-354.

[119] Karl-Heinz Bockstiegel, *Arbitration and Adjudication Regarding Activities in Outer Space*, 6(1) J. OF SPACE L. 7 (Spring 1978). *See, The Convention for the Establishment of a European Space Agency*, 14 I.L.M. 864-908 (May 30, 1975).

than manage crisis situations. The Soviet Union did not provide an early warning that their satellite was going to crash into Canadian territory, although it is questionable whether the location of the crash was predicted with any useful degree of accuracy. Neither did it provide full answers to questions concerning the nature of *Kosmos 954* when the Canadian authorities were attempting to ascertain the potential damage caused.[120] Although this was against the spirit of co-operation which exists in the Liability Convention, that treaty does not stipulate that such details should be provided, its purpose is to regulate damage claims *post hoc*, rather than govern crisis situations.

Article XXI of the Liability Convention addresses instances wherein the crash landing of a space object "presents a large-scale danger to human life or seriously interferes with the living conditions of the population or the functioning of vital centres." In such instances all parties to the treaty, especially the state to whom the object is registered, shall "examine the possibility of rendering appropriate and rapid assistance." Due to the geographic nature of the territory in which the fragments of *Kosmos 954* landed, there was no large scale danger, therefore Canada did not have a legal basis upon which press the U.S.S.R. into providing information pertaining to the satellite, further the weak wording of the treaty would only have required the U.S.S.R. to examine its position. Canada's terse diplomatic language suggests a large degree of irritation with the U.S.S.R. regarding their release of information.[121]

In relation to space debris the case of *Kosmos 954* has implications for whether debris can be considered as 'space objects' in the legal sense of the term. The claim for damages by the Canadian government, and the subsequent payment by the Soviet Union, reveals that both parties considered the constituent parts of the satellite to be 'space objects' and therefore legally the possession, and responsibility, of the U.S.S.R. Although the legal status of debris remains a subject of debate between schol-

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[120] *Canada: Claim Against the Union of Soviet Socialist Republics*, *supra* note 36, at 902-905.

[121] *Id.* at 913-915 & 917-919.

ars,[122] this practical instance strongly suggests that states remain liable for the resulting debris, when space objects fragment.

The fourth space treaty, which had been completed when *Kosmos 954* crashed, was the Registration Convention.[123] This treaty has little applicability to an instance of a crashed space object. However, Article VI creates a legal basis upon which the United States provided assistance to Canada, as it commits signatory states, especially those with space monitoring facilities, to assist in the identification of an object which has landed within sovereign territory. However, it appears more plausible to provide an account of the U.S.A.'s actions in relation to the close relationship between those two states, rather than an instance of space law influencing the actions of states.

The *Kosmos 954* incident highlighted a central weakness in the Registration Convention, the Soviet Union, for strategic reasons, provided a misleading, or at a minimum an incomplete, description in the record of objects launched into space. This was not a factor in the legal considerations arising from the crash of the satellite, however it provides an example of the difficulties associated with the register. The intention of the treaty was to create a mechanism by which orbits could be managed, through the maintenance of a database of all objects placed into near Earth space. However, because it is incomplete, and as *Kosmos 954* demonstrates contains inaccurate information, it has become virtually useless and 'serves little practical or legal purpose'.[124]

The lack of importance associated with the Registration Convention is reflected in the number of states which have rati-

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[122] Lubos Perek, *Managing Issues Concerning Space Debris*, *in* PROCEEDINGS OF THE 4TH EUROPEAN CONFERENCE ON SPACE DEBRIS 589 (Darmstadt, Germany: European Space Agency, 2005).

[123] Registration Convention, *supra* note 9.

[124] Nicholas L. Johnson, *The Earth Satellite Population: Official Growth and Constituents*, *in* JOHN A. SIMPSON, PRESERVATION OF NEAR EARTH SPACE FOR FUTURE GENERATIONS 9 (Cambridge University Press, 1994). Despite its extremely limited utility, the registry is available on the web. *See* United Nations Office for Outer Space Affairs, *Online Index of Objects Launched into Outer Space*, www.unoosa.org/oosa/osoindex.html (last visited Nov. 13, 2008) [hereinafter *Online Index*].

fied it, only forty-six.[125] Lubos Perek argues that a more holistic acceptance of the treaty would strengthen it;[126] certainly it is the case that a complete and accurate database would only be possible if states supplied complete and accurate information. Such a complete register would be beneficial in managing such problem as space debris and nuclear power sources crashing to the Earth.

CONCLUSIONS

In 1978 Karl-Heinz Bockstiegel noted that space law was currently insufficiently equipped to resolve disputes, although they were seemingly inevitable.[127] In the nearly three decades which have passed little has occurred to make space law better able to resolve issues of liability, whilst the presence of debris has increased the possibility that it may be required to address such issues.

There is a clear differentiation to be highlighted concerning *Kosmos 954* and potential instances of damage caused by debris, specifically the location in which damage occurs. The Soviet satellite damaged Canadian property on the surface of the planet whereas the primary, almost exclusive, threat of debris exists in orbit. As such a differing level of liability applies, therefore a direct comparison is not possible. However, as *Kosmos 954* remains the most useful instance in which the provisions of space law and specifically the Liability Convention have been tested, it provides a valuable insight to the mechanics of liability and their application, revealing the institutional framework within which a claim under the Liability Convention would be conducted.

The Liability Convention is designed to restore the situation which existed before damage occurred. This restorative intention means that the treaty does not have the purpose of providing immediate assistance to resolve an incident in which li-

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[125] *Online Index*, *supra* note 124.

[126] Lubos Perek, *Space Debris Mitigation and Prevention: How to Build a Stronger International Regime*, 2(2) ASTROPOLITICS 223 (Summer 2004).

[127] Bockstiegel, *supra* note 119.

ability arises. This is unsurprising as the document is intended to present a legal remedy. Yet in focusing upon restitution, the treaty negates the area in which the launching state could be of most assistance, remedying the immediate difficulties involved in a crisis situation. The most useful assistance which the U.S.S.R. could have given Canada was not the payment of damages, it would have been prior warning that *Kosmos 954* was going to enter Canadian territory, then supplying specific information as to radioactive nature of the satellite. The Liability Convention is notably silent concerning the immediate assistance which should be provided to a state that is subject to a crash, other than when the incident has produced extreme circumstances.[128]

When the Liability Convention was completed it did not provide legal remedies for all possible scenarios, "a number of relatively exotic questions" remained unanswered.[129] As orbital debris had not been identified as a significant problem when the treaty was written, it is a question which although the treaty's principles are applicable they do not directly address. This limits the relevance of the treaty, as debris is the most significant current threat poses to space based activities.

Gorove believes that the case of *Kosmos 954* brought to the fore the "uncertainties and inadequacies" of the legal framework within which such situations would be addressed in the future.[130] A similar situation now applies when considering potential claims for compensation due to space debris; the Liability Convention is untested in its capacity to resolve such an incident. In the case of *Kosmos 954* the treaty operated effectively, whilst it also demonstrated areas in which it created legal uncertainty.

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[128] Liability Convention, *supra* note 5, at art XXI (extreme circumstances, and the appropriate response, are considered).

[129] Such an example could be injuries sustained in space or on a celestial body. It was accepted at the time that in the future a new specific treaty may be required to resolve such situations. Reis, *supra* note 96, at 128.

[130] Gorove, *supra* note 60, at 146.

If Leonid Sedov's claim, that *Kosmos 954* was struck by another craft is correct this could have dramatically altered the question of liability. If the other craft was not of Soviet origin, the state which launched it may have been at fault for the entire sequence of events. Such a scenario would have certainly made the legal proceeding far more complex. As the Soviet government in effect accepted liability by paying damages to Canada, there are three possible explanations; firstly *Kosmos 954* was struck by another craft and that craft was also of Soviet origin. Alternatively, it may be that the Soviet Union ultimately decided that there had not been a collision with another craft. Finally, it is possible that the U.S.S.R. could not prove the origin of the piece of debris which struck *Kosmos 954*, due the debris being small and untrackable. Therefore, it was unable to pursue the relevant launching state for compensation. Concerning *Kosmos 954* it is improbable that the U.S.S.R. believed that debris from another country was responsible for the satellite's demise, but decided to pay compensation regardless. Yet given the difficulty of accurately ascertaining what occurs in orbit this would be a probable scenario for future instances of debris impact.

The *Kosmos 954* incident revealed a great deal concerning the structures and provision created by the Liability Convention, these are relevant to potential future claims for damages caused by space debris. However, a key difference between the two cases does not relate to legal matters, rather it is the political climate during the era. The Cold War is a central feature in the narrative of *Kosmos 954*. The satellite was used for the purpose of spying upon American maritime activities, as such it was partially the political situation which lead to a nuclear powered satellite operating in such a low, and therefore dangerous, orbit. The Cold War climate also had an enormous influence upon the recovery operation. The Canadian government's position was that it requested information concerning the design of the satellite in order to conduct an effective operation. Whilst the same questions, to the Soviet government, were interpreted as inquiries not relevant to the recovery operation, but as an effort to extract strategically useful information.

Thus, the crash of *Kosmos 954* not only provides the most thorough examination of the Liability Convention available, it also provides, in microcosm, a demonstration of the effect of Cold War suspicions upon relations between Eastern and Western states. If the assertions that the two governments made are accepted as true reflections of their positions, then perfectly reasonable requests for technical information were interpreted as an effort to gain a strategic advantage. Given this global atmosphere of suspicion, it is perhaps notable that the legal claim was resolved without recourse to a panel of arbitration.

The Cold War tension defined the approach taken towards resolving problems which arose in the case of *Kosmos 954*, both in the initial Soviet refusal to acknowledge that there was a difficulty with the satellite, and the subsequent difficulties in the recovery operation. Indeed the geo-political climate has a far greater explanatory role concerning the incident than deficiencies within the Liability Convention itself.

Therefore, when considering the role of the treaty in reference to potential problems, which may arise in future, it is firstly important to note the enormous re-orientation there has been in global politics. Further, when considering instances involving orbital debris, it should be remembered that there is common interest in avoiding problems, and sharing information, as opposed to the zero-sum interactions relating to crashed spy satellites.

The two most significant problems which hampered the Canadian government, in the instance of *Kosmos 954*, were a lack of knowledge concerning where the satellite was to land, and the lack of cooperation in the recovery operation. In the instance of space debris, it is improbable that such problems will occur. There has been a notably cooperative international response to the problem.[131] As such, the management of the debris

[131] The IADC provides an institutional framework within which policy towards debris can be coordinated between several states. The member states have written guidelines for all states in order to commonly address the problem. It can therefore be considered to be one which has address through international cooperation.

problem has been characterised by dangers being identified through the sharing of tracking information, such cooperation can be reasonably expected to characterise initial responses to impacts.

The question remains as to whether cases of liability would be resolved in a similar spirit. An initial difference between *Kosmos 954*, and a potential incident of a satellite being damaged by debris is the availability of information. When an object crashed into Canadian territory there was no doubt that it was the Soviet craft *Kosmos 954*; should a satellite be damaged in orbit there would be far less certainty. The most probable scenario is that the debris responsible would be too small to be tracked, therefore it would be impossible for liability to be proved. Should the origin of the debris be known, the situation would still not be simply resolved. For, when damage occurs in space, the provisions of the Liability Convention require fault to be shown, this would be difficult to prove in the instance of a dead satellite, spent rocket stage or stray bolt. Also in the absence of access to the physical evidence the possibility remains that a state may simply deny that the object involved was its property.

The treaties compromising space law were written before space debris was identified as a serious problem, as such they are not easily applied to the issue. This piece has examined whether the Liability Convention could address future incidents concerning debris. The case of *Kosmos 954* revealed the manner in which the treaty would attempt to resolve problems, it was found that the there are provisions of the treaty are not easily applicable to the problem of debris. Firstly, the issue of which state launched an object which has caused damage is particularly difficult to ascertain when considering small pieces of debris. Secondly, the Liability Convention when addressing incidents of damage in space requires fault to be shown, it is difficult to conceptualise how pieces of debris which are in orbit, as a by-product of space activities, could be considered to exist because of fault by states. Finally, the treaty aims to be restorative as such it is seeking to address the situation after damage

has occurred, whereas the current need with reference to debris is to reduce the possibility of incidents occurring, rather than managing a situation after damage has occurred.

FY-1C AND *USA-193* ASAT INTERCEPTS: AN ASSESSMENT OF LEGAL OBLIGATIONS UNDER ARTICLE IX OF THE OUTER SPACE TREATY

Michael C. Mineiro[*]

I. INTRODUCTION

On January 11, 2007[1] (EST), China performed a successful anti-satellite (ASAT) missile test using a kinetic kill vehicle launched from the Xichang Satellite Launch Center (XSLC), a facility in Sichuan Province, China.[2] The kinetic kill vehicle destroyed an aging, but functioning, Chinese weather satellite, the *Feng Yun 1C* (*FY-1C*), in polar orbit at an altitude of approximately 537 miles.[3] This was the first such destruction of a satellite since the kinetic ASAT missile tests conducted during the Cold War by the United States and the Soviet Union.[4]

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[1] The test occurred on January 11, 2008 in the United States, but because of the difference in local times the event occurred on January 12, 2008 in China.

[2] Craig Covault, *Chinese Test Anti-Satellite Weapon*, AVIATION WEEK, Jan. 17, 2007, http://www.aviationweek.com/aw/generic/story_channel.jsp?channel=space&id=news/CHI01177.xml; Craig Covault, *China's ASAT Test Will Intensify U.S.-Chinese Standoff*, AVIATION WEEK, Jan. 24, 2007), http://www.aviationweek.com/aw/generic/story_generic.jsp?channel=awst&id=news/aw012207p2.xml.

[3] T.S. Kelso, *Analysis of the 2007 Chinese ASAT Test and the Impact of its Debris on the Space Environment*, 2007 AMOS CONFERENCE 321 (2007), http://www.centerforspace.com/downloads/files/pubs/AMOS-2007.pdf.

[4] Shirley Kan, *China's Anti-Satellite Weapon Test*, CRS REPORT FOR CONGRESS, Apr. 23, 2006, http://fpc.state.gov/documents/organization/84322.pdf.

One year later, on February 14, 2008, the United States announced it was planning to destroy *USA-193* (a.k.a. *NROL-21*), a derelict satellite that was decaying out of orbit and falling back to Earth.[5] On February 21, 2008, the United States successfully destroyed *USA-193* at an altitude of approximately 133 miles with a kinetic ASAT missile.[6]

In the wake of these State actions, the question is raised: What, if any, affirmative legal obligations does Article IX of the Outer Space Treaty[7] impose on States Party to the Treaty planning to or carrying out ASAT (anti-satellite) activities or experiments in outer space? Did China and the United States act in accordance with Article IX?

In the following article, these questions are assessed under a methodology of treaty interpretation consistent with the Vienna Convention on the Law of the Treaties. The negotiating history and historical contest of Article IX and the Outer Space Treaty are examined. Article IX is assessed within a matrix of principles and corresponding articulations of affirmative obligations. This assessment is then applied to the facts publicly known regarding the *FY-1C* and *USA-193* intercept for the purpose of concluding whether or not Article IX obligations were applicable and if applicable whether they were fulfilled. Finally, conclusions are reached as to the Article IX legality of the *FY-1C* and *USA-193* intercepts and discussed within the greater context of international law and governance.

II. TREATY INTERPRETATION METHODOLOGY

All too often analyses of treaty obligations are written without first defining the methodology of treaty interpretation adopted for the analyses. This can result in the discussion be-

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[5] U.S. Department of Defense News Transcript, *DoD News Briefing with Deputy National Security Advisor Jeffrey, Gen. Cartwright and NASA Administrator Griffin*, Feb. 14, 2008 [hereinafter *DoD News Briefing*].

[6] Thom Shanker, *Missile Strikes a Spy Satellite Falling From Its Orbit*, NEW YORK TIMES (Feb. 21, 2008), http://www.nytimes.com/2008/02/21/us/21satellite.html?ex=1361336400&en=ea5702ff269483cc&ei=5088&partner=rssnyt&emc=rss.

[7] Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and other Celestial Bodies, Jan. 27, 1967, 18 U.S.T. 2410, 610 U.N.T.S. 205 [hereinafter Outer Space Treaty].

ing directed not to the question of interpretation methodology and law, but instead on presumptions formulated from premises neither party articulates. Let us avoid this pitfall by defining and adopting a method of treaty interpretation.

Numerous methods of treaty interpretation exist. Common methodologies include the textual, subjective, and teleological approaches, and variants thereof.[8] Often no single approach is adopted, but rather an amalgam of two or more approaches. Common among these approaches is the shared premise that "the validity of an international agreement rests solely on the will or consent of the parties to be bound and, thereof, treaty interpretation is the process of attempting to establish the context of the consent."[9]

The standard method of treaty interpretation, as adopted by the International Court of Justice, is articulated in Articles 31 and 32 of the Vienna Convention on the Law of the Treaties.[10] Articles 31 and 32 are considered by the Court as having attained the status of customary international law.[11] Article 31, General Rule of Interpretation, states:

> 1. A treaty shall be interpreted in good faith in accordance with the ordinary meaning to be given to the terms of the treaty in their context and in the light of its object and purpose.
>
> 2. The context for the purpose of the interpretation of a treaty shall comprise, in addition to the text, including its preamble and annexes:
>
> (a) any agreement relating to the treaty which was made between all the parties in connection with the conclusion of the treaty;

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[8] Francis Jacobs, *Varieties of Approach to Treaty Interpretation: With special reference to the Draft Convention on the Law of Treaties before the Vienna Diplomatic Conference*, 18 INT. AND COMP. L. QUART. 318, 319 (1969).

[9] Kenneth Vandervelde, *Treaty Interpretation from a Negotiator's Perspective*, 21 VAND. J. TRANSNAT'L L. 281, 287 (1988).

[10] Vienna Convention on the Law of Treaties, May 23, 1969, 1155 U.N.T.S. 331.

[11] ANTHONY AUST, MODERN TREATY LAW AND PRACTICE 232 (2d ed., (Cambridge University Press, 2d ed., 2007).

(b) any instrument which was made by one or more parties in connection with the conclusion of the treaty and accepted by the other parties as an instrument related to the treaty.

3. There shall be taken into account, together with the context:

(a) any subsequent agreement between the parties regarding the interpretation of the treaty or the application of its provisions;

(b) any subsequent practice in the application of the treaty which establishes the agreement of the parties regarding its interpretation;

(c) any relevant rules of international law applicable in the relations between the parties.

4. A special meaning shall be given to a term if it is established that the parties so intended.

Article 32, Supplementary Means of Interpretation, states:

Recourse may be had to supplementary means of interpretation, including the preparatory work of the treaty and the circumstances of its conclusion, in order to confirm the meaning resulting from the application of article 31, or to determine the meaning when the interpretation according to article 31:

(a) leaves the meaning ambiguous or obscure; or

(b) leads to a result which is manifestly absurd or unreasonable.

As articulated in Article 31(1), the Vienna Convention adopts first and foremost an objective textual approach based on the premise of a neutral third party observer. Although focused on the textual approach, the Convention is not strictly textual as both subjective and teleological approaches are included within the methodology of the articles.[12] Inherent in the objective textual approach is the recognition that "the primary aim of international law is not to provide the ideal method of resolving

[12] Jacobs, *supra* note 8.